



SYMBIOSIS INSTITUTE OF TECHNOLOGY (SIT)

Constituent of Symbiosis International (Deemed University), Pune

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Re-Accredited by NAAC with 'A++' Grade

ASSIGNMENT No: 14

TITLE: Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.



Exercise

1. Create an ATM program handling invalid withdrawal amount using try-catch

```
1 import java.util.Scanner;
2 public class Shop {
3     public static void main(String[] args) {
4         Scanner sc = new Scanner(System.in);
5         try {
6             System.out.print("Enter product quantity: ");
7             int quantity = sc.nextInt();
8             if (quantity <= 0) {
9                 throw new IllegalArgumentException("Invalid product quantity.");
10            }
11            System.out.println("Product quantity: " + quantity);
12            System.out.println("Order placed successfully.");
13        }
14        catch (IllegalArgumentException e) {
15            System.out.println("Error: " + e.getMessage());
16        }
17        sc.close();
18    }
19 }
20
```

(Shop) #2

Ln 20, Col 1 Spaces: 4 UTF-8 CRLF { } Java Go Live

Windows taskbar: Search, Task View, File Explorer, Settings, Dev C++, 20% battery, 14:52 20-08-2026

```
Problems 1 Output Debug Console Terminal Ports

PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14> cd assign14
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> javac Shop.java
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> java Shop
Enter product quantity: 0
Error: Invalid product quantity.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14>
```

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14> cd assign14
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> java Shop
Enter product quantity: 10
Product quantity: 10
Order placed successfully.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14>
```



2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
1 import java.util.Scanner;
2 public class Shop {
3     Run main | Debug main
4     public static void main(String[] args) {
5         Scanner sc = new Scanner(System.in);
6         try {
7             System.out.print("Enter product quantity: ");
8             int quantity = sc.nextInt();
9             if (quantity <= 0) {
10                 throw new IllegalArgumentException("Invalid product quantity.");
11             }
12             System.out.println("Product quantity: " + quantity);
13             System.out.println("Order placed successfully.");
14         } catch (IllegalArgumentException e) {
15             System.out.println("Error: " + e.getMessage());
16         }
17         sc.close();
18     }
19 }
20
```

(Shop) #2 Ln 20, Col 1 Spaces: 4 UTF-8 CRLF { } Java Go Live

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14> cd assign14
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> javac Shop.java
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> java Shop
Enter product quantity: 0
Error: Invalid product quantity.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> |
```

```
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14> cd assign14
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> java Shop
Enter product quantity: 10
Product quantity: 10
Order placed successfully.
PS C:\Users\pdave\Downloads\Java sem 3\experiments\Assignment 14\assign14> |
```